Q1 - Speed of sound and compression: Recall that the speed of sound for anything is:

$$\frac{dP}{d\rho} = c_s^2 \tag{1}$$

where dP is the change in pressure, $d\rho$ is the change in density, and c_s is the speed of sound. First, show that

$$d\rho = \frac{dP}{c_s^2} \tag{2}$$

This tells us that if we apply extra pressure to something (dP) that the density ($d\rho$) changes by that pressure divided by the speed of sound squared. It may seem like solids don't shrink when we push on them, but they do and the speed of sound in that solid tells us how much. The speed of sound is effectively a stiffness - a higher speed of sound means a smaller change in density for a given applied pressure.

The speed of sound in sea water is 1500 m/s. What is the pressure at the bottom of the Marianas Trench, 11 km below sea level? What is the change in density between the surface and the bottom? What is the *fractional* change in density? And is the water more dense or less dense in the ocean depths? You may assume the density of water is 1000 kg/m^3. The true answer is about 5%, so our estimate is very close!

Q2 - Two strings are tied together. String two has twice the mass per unit length compared to string one. Someone at the end of string one sends a wave down it. What are the amplitudes of the reflected and transmitted waves, relative to the incoming wave, when the incoming wave hits the junction?

Q3 - A violin body has a volume of about 1850 cubic cm. Its f-holes are tuned so the Helmholtz resonance is at 330 Hz. Typical dimensions are about 8mm wide and 75mm long. What is the effective length of the resonating air plug? You should find it's comparable to the width of the f-hole. Remember that the resonant frequency of a Helmholtz resonator is

$$\nu = \frac{v}{2\pi}\sqrt{\frac{A}{LV}} \tag{3}$$

Q4 - As mentioned in class, a cracked window in a moving car often leads to a thumping Helmholtz resonance. Would you exect the frequency of the thumping to depend on the speed of the car?

Q5 - You see a flash of lightning. Exactly 10 seconds later, you hear thunder. How far away

was the lightning if the day was very hot (+35 C)? Very cold (-10 C)?